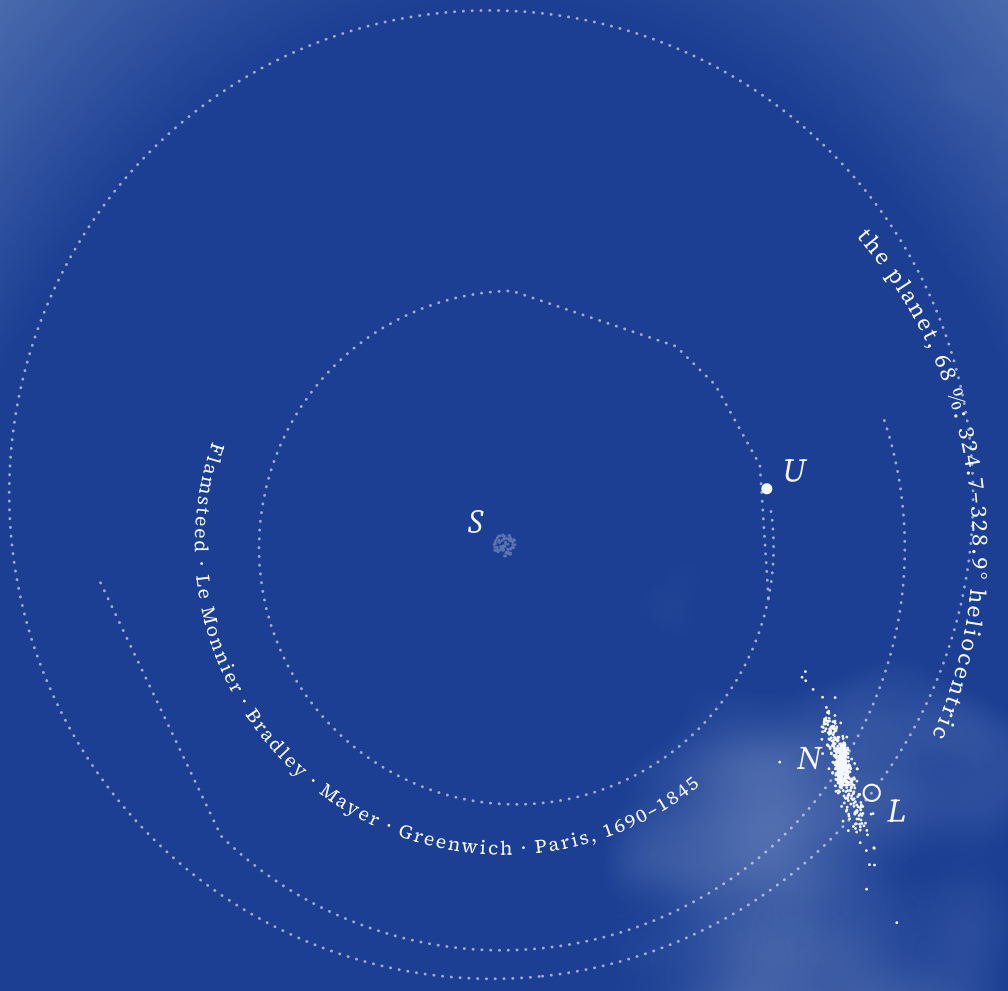
PLATE I.



Refitting Le Verrier's prediction of Neptune

A pre-registered Bayesian reconstruction from the 279 observations of Uranus he published in 1846

Atishay Pahadiya · 2026 · version 10.5281/zenodo.24044792

PLATE I. The planet on Galle's night, 23 September 1846, 23:00 TDB: heliocentric positions projected on the ecliptic of date, seen from the north with the equinox to the right; to scale. *S* Sun. *U* Uranus (median of the fitted positions) on its dotted path from JPL Horizons, lettered with the sources of Table 1. Stipple: 600 of the 6000 posterior samples of the planet (M6c replacement posterior); heliocentric longitude 68 %: 324.7–328.9°. *N* Neptune (JPL Horizons). *L* Le Verrier's planet from his published elements (Krajnović 2016, after Grosser 1962).

§ 1

Abstract

We refit the 279 observations of Uranus (529 measured coordinates, 1690–1845) that Le Verrier published in 1846, with a perturbed-orbit model that is never given Neptune, and sample the posterior by partitioned nested sampling under a plan committed before the run. On the night Galle found Neptune, 23 September 1846, we find the unseen planet's geocentric longitude (ecliptic of date) at 325.8° (68 %: 323.6–327.8°; 95 %: 320.2–330.4°, provisional). Neptune's longitude was 325.87°, 0.06° from the median. The fitted mass is 17.3 Earth masses (68 %: 16.3–18.4; Neptune 17.15) and the semi-major axis 30.2 AU (29.6–30.9). Le Verrier's published orbit gives a geocentric longitude of 324.97° that night, inside our 68 % range. In heliocentric longitude (ecliptic of date) his planet lies outside our 95 % range at the conjunction of 1821 (§ 4.2) and stays within the 68 % band only from 1835.75 to 1856.50 (§ 4.3).

§ 2

Data: Le Verrier’s observations of Uranus

Table 77 of Le Verrier's memoir (Le Verrier 1846) lists 279 observations of Uranus from 1690 to 1845 on 8 printed pages, each beside the position he computed. We transcribed it in two passes; they disagreed on 14 cells, each resolved against the scan, and 7 misprints in the print are logged with the argument for each. The transcription was machine-assisted, not done by an independent human. 28 rows (336 cells) were then checked against the scan by machine and by hand, with 0 discrepancies; the other 251 rows have had no human check. The transcription is published as a dataset (Pahadiya 2026).

Table 1 groups the observations by source: the four observers before Herschel, whom the error model pools into one group, and the three modern series that are its other groups. The last column is each source's median distance from JPL's Uranus taken through our observation model (§ 3.3). These residuals set the error level of each group in the likelihood; sensitivity run S2 replaces them with levels estimated from the data alone (§ 5).

Source	Observatory	Years	n	RA	Dec	|O − JPL|
Flamsteed	Greenwich	1690–1715	5	5	5	5.37″
Le Monnier	Paris	1750–1771	12	12	12	3.38″
Bradley	Greenwich	1753	1	1	0	1.07″
Mayer	Göttingen	1756	1	1	1	1.41″
Greenwich series	Greenwich	1781–1830	87	87	80	1.81″
Paris series	Paris	1801–1827	107	107	95	1.93″
Paris series	Paris	1835–1845	66	66	57	1.02″
All sources		1690–1845	279	279	250	1.66″

28 rows checked by hand against the scan (A6a). CC0 1.0, doi:10.5281/zenodo.24044792.

TABLE 1. Le Verrier's Table 77 by source. n: observations (rows). RA, Dec: measured coordinates. |O − JPL|: median absolute on-sky residual against JPL's Uranus at the apparent place of date, in arcseconds (right ascension as Δα cos δ). Built from the published transcription; the per-group medians reproduce the committed error model.

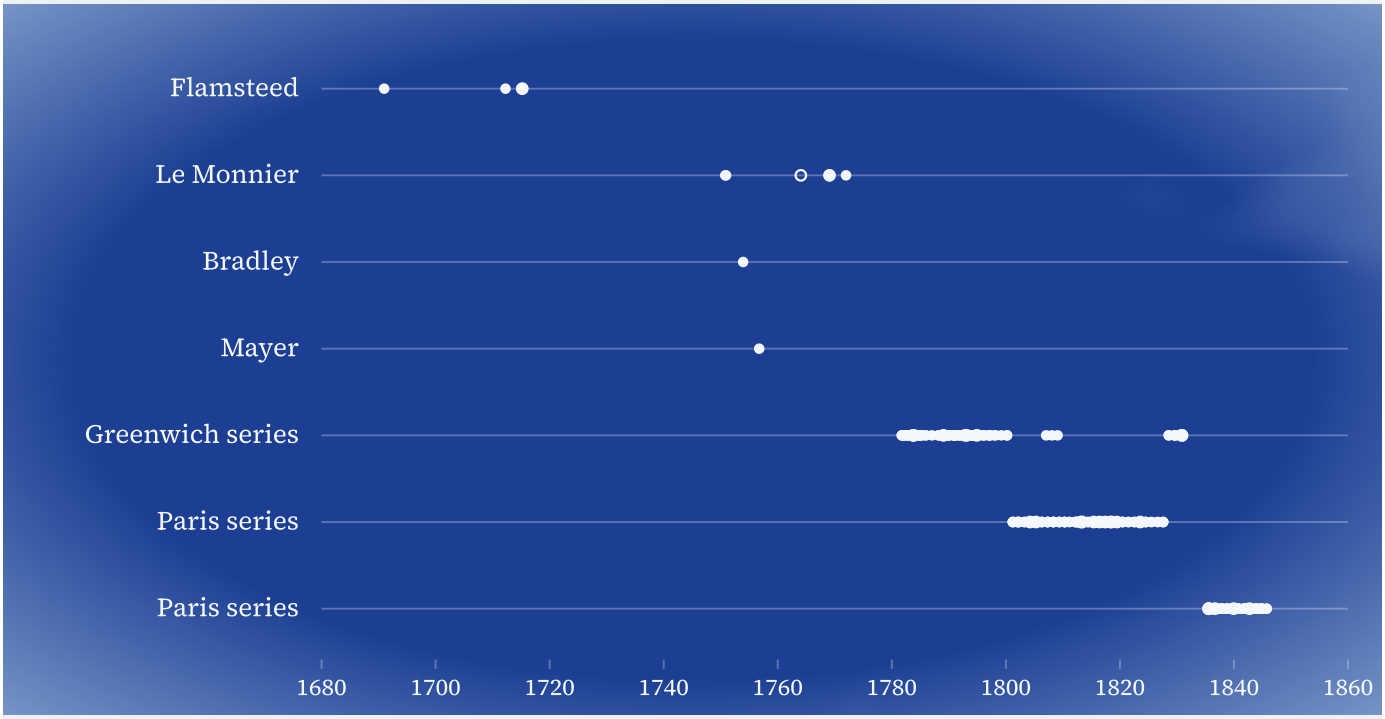


FIG. 1. The 279 observations of Table 77 by source and year, one dot per observation; the 28 rows checked by hand are open circles. Hover for the date and the residuals against JPL.

§ 3

Method

§ 3.1 Pre-registration

Every threshold, reading rule and run setting below was committed to the repository before the run it governs, and each commit's first push is a public record. Fig. 2 places each run set on its own launch time. Times are PST, from git, GitHub's activity API and the run logs.

The headline fit's plan was committed at 15:13:24 on 7 Nov and its run launched 5 s later, at 15:13:29. The plan was first pushed at 15:30:17, 16 min 48 s after launch: public 2 h 50 min before the run finished (18:20:31) and 2 h 54 min before the first summary of its result (18:24:43). The logs cannot rule out that one of its result files was opened by hand during the run. Every later plan was pushed before its runs launched, by 15 s to 4 h 2 min.

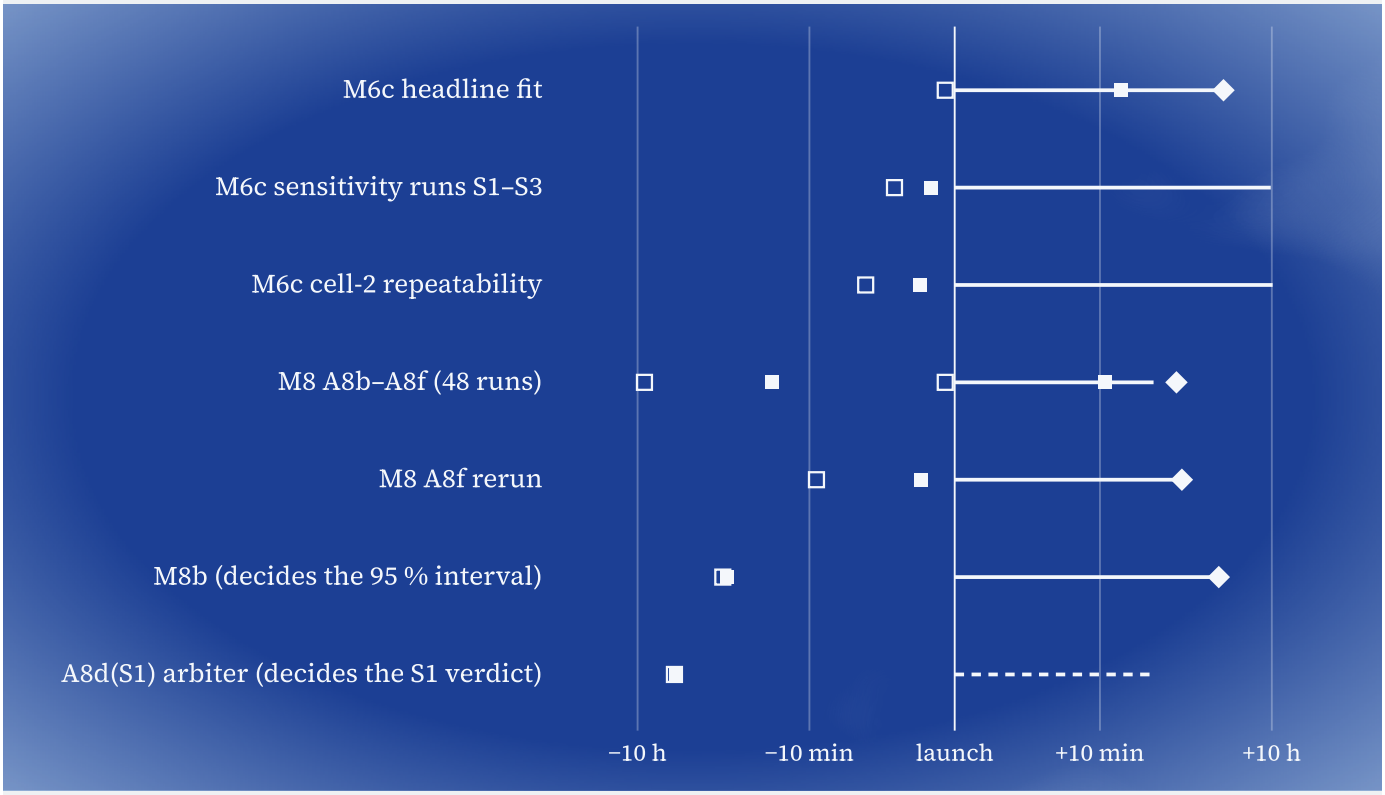


FIG. 2. When each plan went public. One row per pre-registered run set, each aligned on its own launch (the vertical line); time on a signed logarithmic scale. Open square: plan committed; filled square: first pushed; bar: the run, launch to finish (dashed while running: A8d(S1) arbiter (decides the S1 verdict)); diamond: first summary of the result. From the repository's timeline file, generated 9 Nov 17:00:38.

§ 3.2 The perturbation Le Verrier inverted

> “It is impossible, I think, to read this letter without being struck with its clearness of explanation, with the writer's extraordinary command, not only of the physical theories of perturbation but also of the geometrical theories of the deduction of orbits from observations, and with his perception that his theory ought to explain all the phenomena of the planet's place”
>
> G. B. Airy, on Le Verrier's letter to him of 28 June 1846, Airy (1846), Monthly Notices of the Royal Astronomical Society, 7, 121; quoted in Krajnović (2016).

Plate III shows the perturbation Le Verrier inverted, computed directly. The Sun and the eight planets are integrated from JPL's states at Herschel's discovery, once with Neptune's mass and once without. *K* marks the heliocentric conjunction of Uranus and Neptune, 28 September 1821, 07:22 TDB (ecliptic of date; JPL's date to 0.2 h). It is Neptune's own conjunction, not § 4.2's *C*, the conjunction with the fitted planet. Before *K* Neptune's pull has a component along Uranus's motion that draws it ahead; from 21 February 1822, 02:14 TDB, 145.8 days after *K*, it holds Uranus back. By 1849.9 Uranus is 180.9″ behind where it would be without Neptune.

PLATE III.

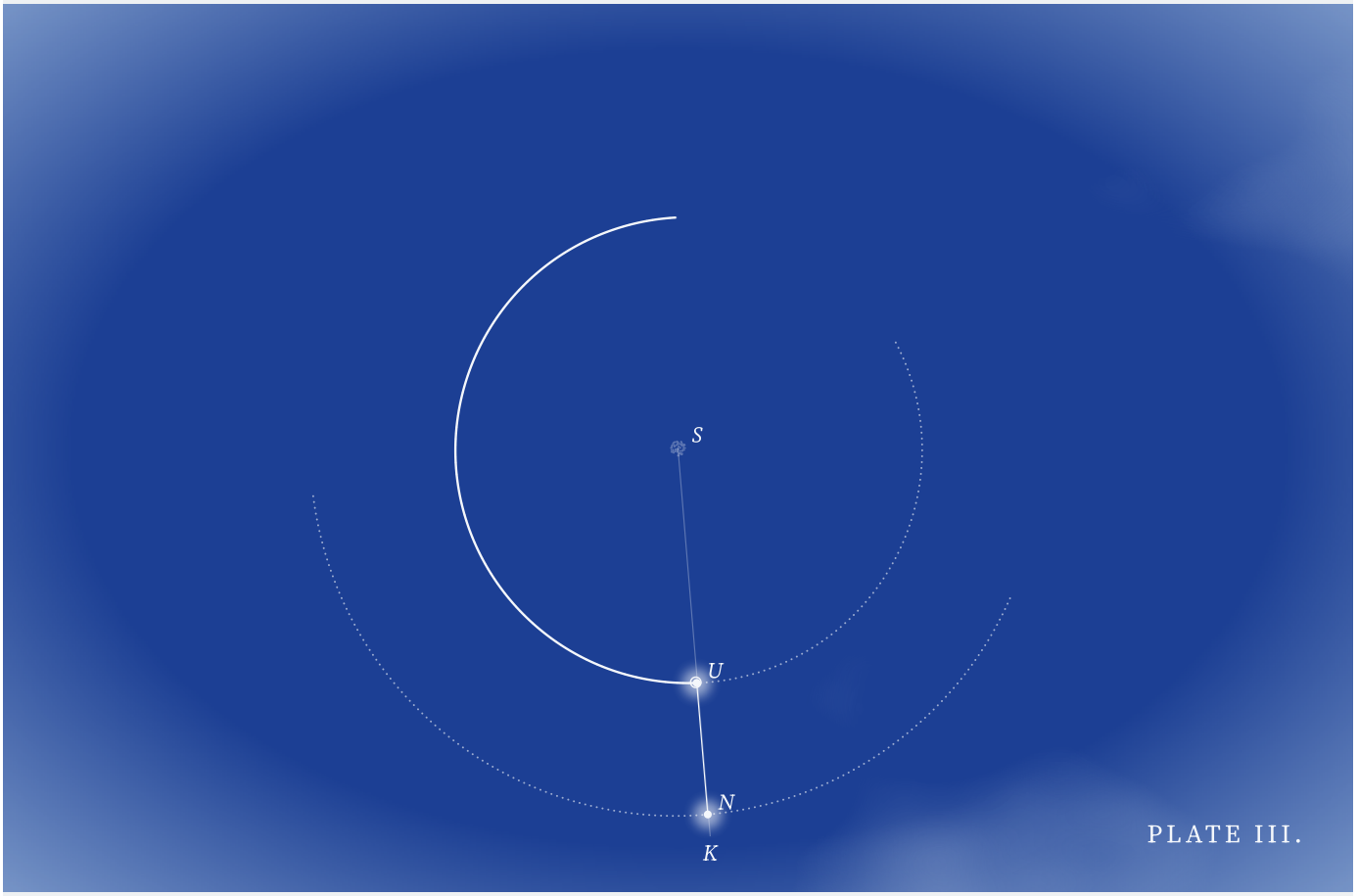


PLATE III. Uranus (*U*) and Neptune (*N*) about the Sun (*S*): heliocentric, ecliptic and mean equinox of J2000, seen from the north with the equinox to the right; to scale, except the open circle (offset ×50). REBOUND 5.2.2, IAS15 (Rein & Liu 2012; Rein & Spiegel 2015): the Sun and eight planets from JPL Horizons states at 13 March 1781, 0h TDB, run with and without Neptune's mass, drawn every 0.1 yr to 1849.90. The line *UN* carries Neptune's pull. *K*: the heliocentric conjunction of Uranus and Neptune in

the run, 28 September 1821, 07:22 TDB (ecliptic of date), 0.2 h from JPL's; drawn at Neptune's place on that date. It is not Plate II's *C*. Horizons check: at 1 January 1846, the last cached epoch (none reaches J1850.0), the run's Uranus differs from Horizons' by 0.009″, the largest difference over all 1236 cached epochs checked.

1821

Uranus is drawn ahead by Neptune's pull, and 15.9″ ahead of its Neptune-free path.

Open circle: Uranus as it would move without Neptune, from the same 1781 state; its offset is drawn 50 times larger than life.

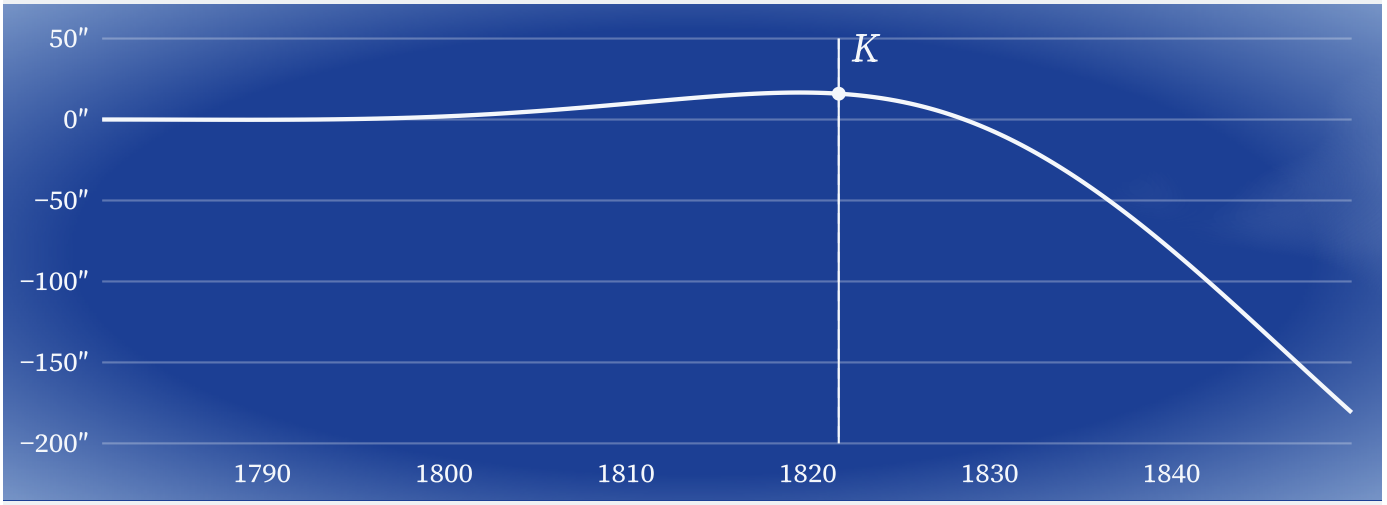


FIG. 3. Uranus's heliocentric longitude (ecliptic J2000) with Neptune minus without, in arcseconds, 1781–1849.9. The cursor follows the plate's year; *K* the conjunction of Uranus and Neptune (*K* in Plate III).

§ 3.3 Observation model, likelihood and priors

Each measured right ascension and declination is predicted by integrating the Sun (with the inner planets folded in), Jupiter, Saturn, Uranus and one unseen planet with our own Wisdom–Holman integrator, then carrying Uranus through light time, annual aberration, precession and nutation (Capitaine et al. 2003) to the apparent place of date, the frame of Table 77. Jupiter and Saturn start from JPL states and are not fitted. Uranus's four in-plane elements are free; its inclination and node are held at their 1781 values, which positions in longitude cannot constrain. The planet has seven parameters: mass, semi-major axis, eccentricity, two orbital angles, and its inclination and node relative to Uranus's plane.

The likelihood is Gaussian in 8 error groups (the ancient observers, Greenwich, Paris 1801–1827 and Paris 1835–1845, each in RA and Dec), with σ from 1.16″ to 5.08″ calibrated against JPL and a free offset per group (prior σ 2–5″). Priors: mass log-uniform over 1–100 Earth masses; semi-major axis log-uniform over 20–60 AU; eccentricity uniform up to 0.3; inclination to Uranus's plane up to 10°, uniform in cos i; the angles uniform. The posterior is sampled with dynesty (Speagle 2020) in 4 cells (20–25 AU; 25–28 AU; 28–60 AU with *e* 0–0.12; 28–60 AU with *e* 0.12–0.3) with 1000 live points (2000 in the first), and the cells are weighted by their evidence. The code path of the fit never reads Neptune; a static check of its call graph and a runtime test confirm it (A6d).

§ 4

Results

§ 4.1 The planet's longitude on Galle's night

> “Le [sic] planète, dont vous avez signalé la position, réellement existe”
>
> “The planet whose position you predicted really exists” (translation: Krajnović (2016))
>
> J. G. Galle to U. J. J. Le Verrier, letter, Berlin, 25 September 1846; quoted in Krajnović (2016).

Every longitude and difference in this section is geocentric, ecliptic of date. On Galle's night, 23 September 1846, we find the planet's longitude at 325.8°, 68 % range 323.6–327.8°. Neptune was at 325.87° (JPL Horizons), 0.06° from the median, 0.03 of the 68 % half-width. The 95 % range, 320.2–330.4°, is provisional: M8b did not meet the replacement rule (std_95_bounds_le_0p4). Le Verrier's published elements, carried to that night (our calculation; Kepler orbit, equinox 1847.0 assumed), give *L* at 324.97°, 0.84° from our median (geocentric), inside the 68 % range.

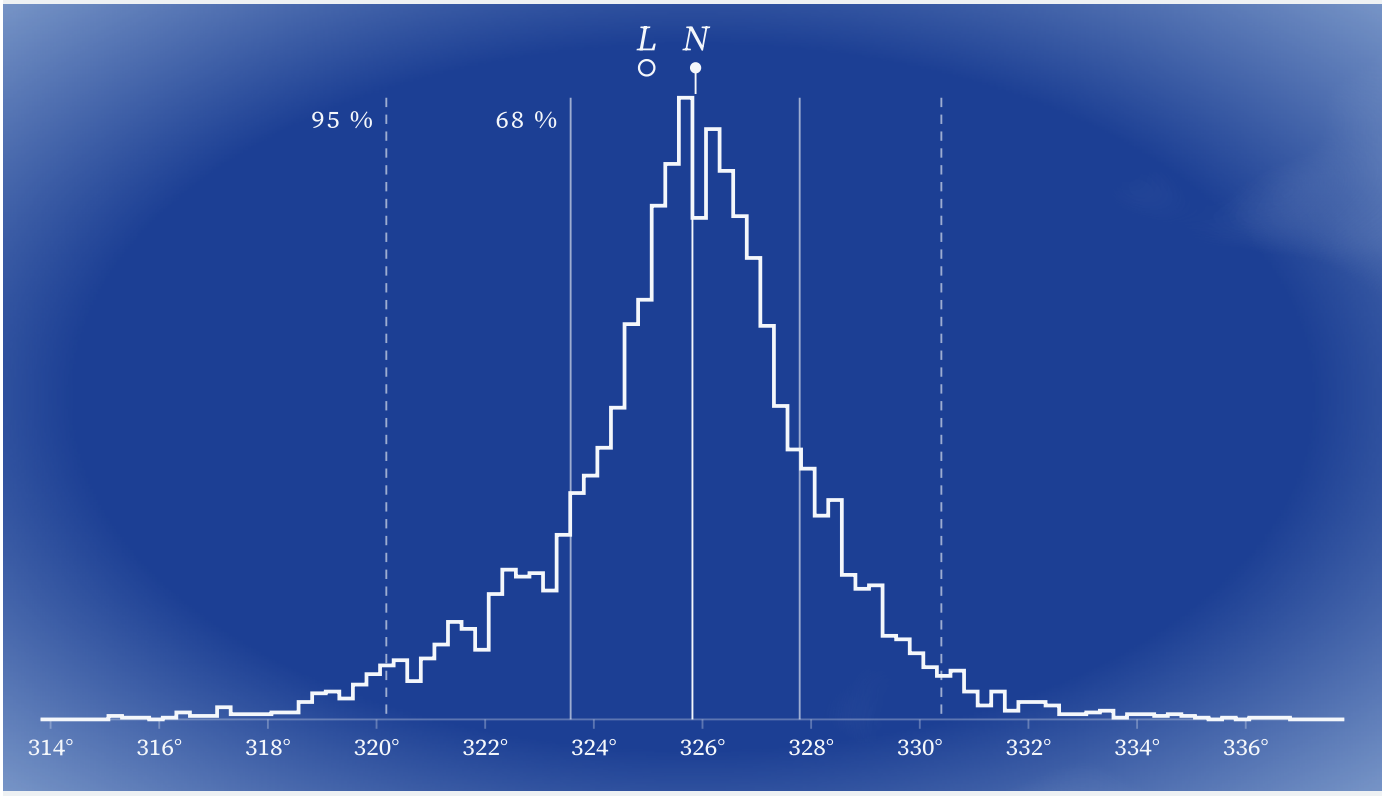


FIG. 4. Posterior of the planet's geocentric longitude on 23 September 1846, ecliptic of date: histogram of the 6000 derived samples of the M6c replacement posterior. Solid lines: median and 68 % range; dashed: the 95 % range (provisional). *N* Neptune (JPL Horizons). *L* Le Verrier's elements carried to that night.

The same posterior gives a mass of 17.3 Earth masses (68 %: 16.3–18.4; Neptune 17.15), a semi-major axis of 30.2 AU (29.6–30.9), osculating in 1781, and an eccentricity below 0.0276 (68 %, one-sided). The inclination to Uranus's plane is dominated by its prior, and we do not report it as a result. The sampler's own systematic on the median is ±0.13°, not folded into the range.

§ 4.2 The conjunction of 1821

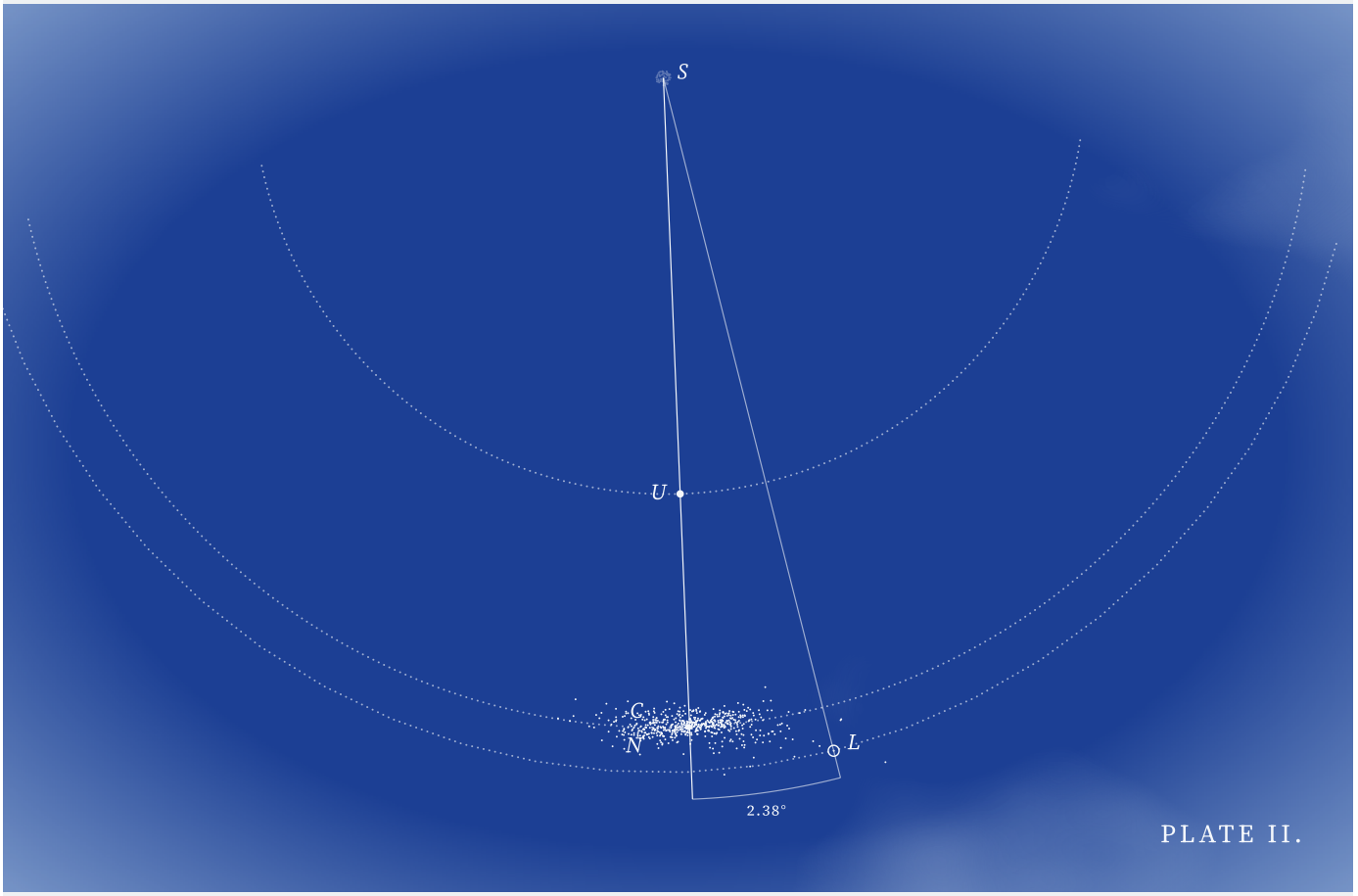


PLATE II. Heliocentric positions on 3 October 1821, 0h TDB, projected on the ecliptic of date, seen from the north with the equinox to the right. Radii to scale; angles about the line *SC* drawn 5 times larger than life (the angle label gives the true value). *S* Sun. *U* Uranus (median of the fitted positions). Stipple: 600 of the 6000 posterior samples of the planet (M6c replacement posterior). *C* their median direction. *N* Neptune (JPL Horizons). *L* Le Verrier's planet from his published elements (Krajnović 2016; Britannica 1911). Dotted: the paths of Uranus and Neptune from JPL Horizons and Le Verrier's ellipse. After-the-fact diagnostic, not pre-registered.

On 3 October 1821, 0h TDB, Uranus (*U*) and the planet were in heliocentric conjunction: the line *SU* produced passes through *C*, the median of the planet's heliocentric longitude, 272.31° (ecliptic of date; 68 %: 271.62–273.02°). Neptune (*N*, JPL) was at 272.29°. Le Verrier's orbit puts his planet at *L*, 274.69°: the angle *CSL* is 2.38°, 3.36 half-widths of our 68 % range, outside the 95 % range.

The date of *C* is the median, over the 6000 posterior samples, of the date on which each sample's planet and Uranus share a heliocentric longitude (ecliptic of date); 68 % of those dates fall within 0.69 yr. The same rule applied to JPL's Uranus and Neptune gives 28 September 1821, 07:22 TDB, 4.7 days earlier, well inside that spread. This comparison was made after the fact; it was not pre-registered.

§ 4.3 When Le Verrier's orbit agrees with the data

> “You are recognised beyond doubt as the real predictor of the planet's place”
>
> G. B. Airy to U. J. J. Le Verrier, private letter, 14 October 1846; quoted in Krajnović (2016).

Hold Le Verrier's published orbit fixed and carry it through time. Every longitude and difference in this section is heliocentric, ecliptic of date. His planet crosses our posterior median once, at 1842.09, and stays inside the 68 % band from 1835.75 to 1856.50. On Galle's night it was 1.0° from Neptune in heliocentric longitude (the 0.84° of § 4.1 is geocentric, and from our median). A search on 23 September 1836 would have been off by 0.8°; one on 23 September 1856, by 3.8°. We did not model the search fields or the star charts.

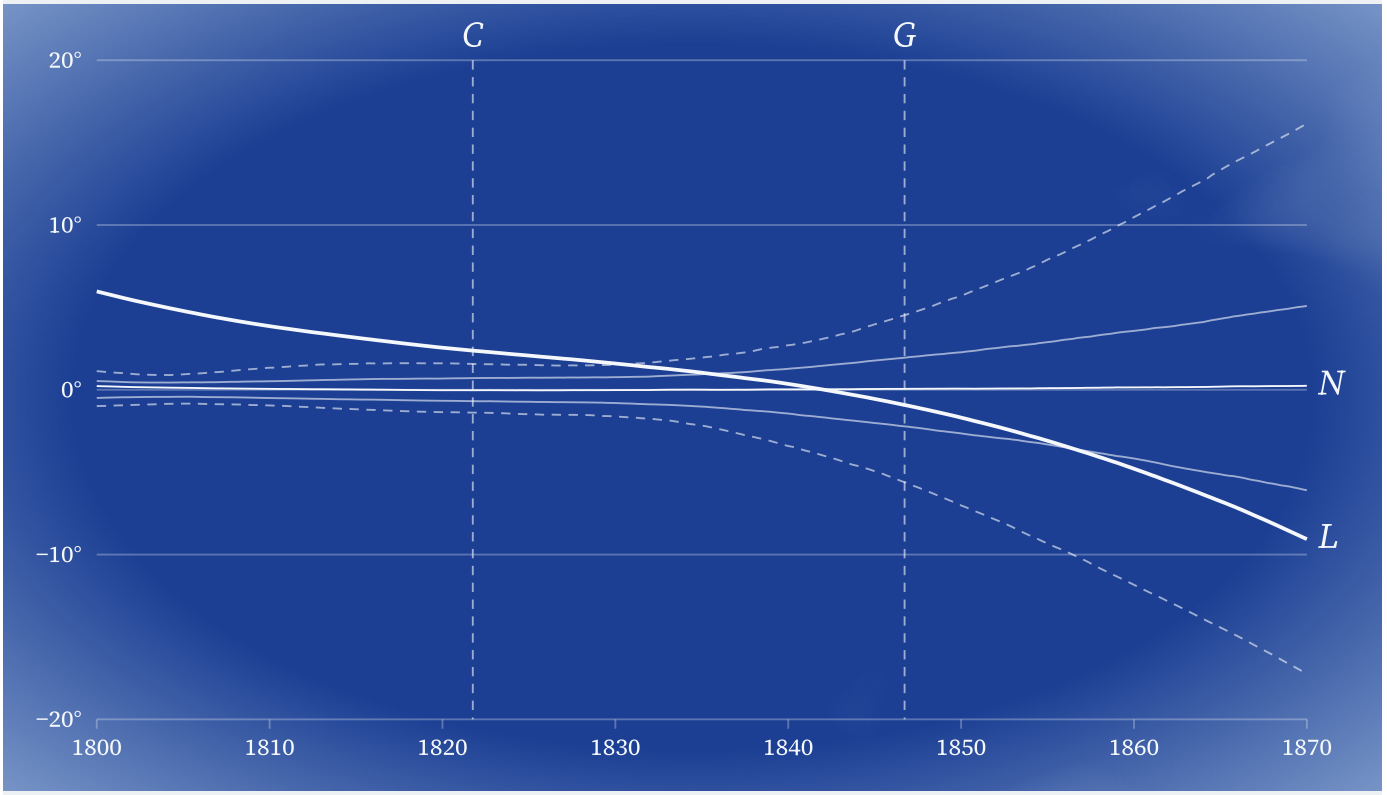


FIG. 5. Heliocentric longitude (ecliptic of date) relative to the posterior median, 1800–1870: the 68 % band (solid edges) and 95 % band (dashed edges) of the M6c replacement posterior; *L* Le Verrier's planet (heavy line); *N* Neptune (JPL Horizons). *C* the conjunction of § 4.2; *G* Galle's night. After-the-fact diagnostic, not pre-registered.

§ 5

Robustness

The first published 68 % range, 323.91–327.39° (3.48° wide), rested on one run of the main cell. Four repeats at the same settings gave widths from 3.23° to 4.57°, a spread 10 times the sampler's own error estimate. Two runs with four times the live points agreed on the width to 1.0 % (rule: within 10 %); under the pre-registered rule their merge replaced that cell, and the range became 323.57–327.79° (4.22° wide).

Three sensitivity runs, with reading rules committed before they ran, test the model's assumptions. With eccentricity allowed up to 0.5 (S1), the published run gives P(a < 25 AU) = 0.001, which the rule reads as "not at all"; M8's faster likelihood gives 0.109, and the verdict is pending a full-likelihood arbiter run. With error levels estimated from the data alone (S2) the result survives: the median moves 0.02 headline half-widths. The evidence favours Student-t errors, $\nu = 4$ (S3), over the Gaussian ones: $\ln B = +13.7 \pm 0.5$. The Gaussian error model was fixed by the pre-registration and stays the headline's, and S3

shows the result is insensitive to that choice: the median moves 0.37 headline half-widths, while S3's 68 % range is 6.33° wide against the headline's 4.22°. Neptune's longitude (JPL) is inside the 68 % range in the headline fit and in all three runs.

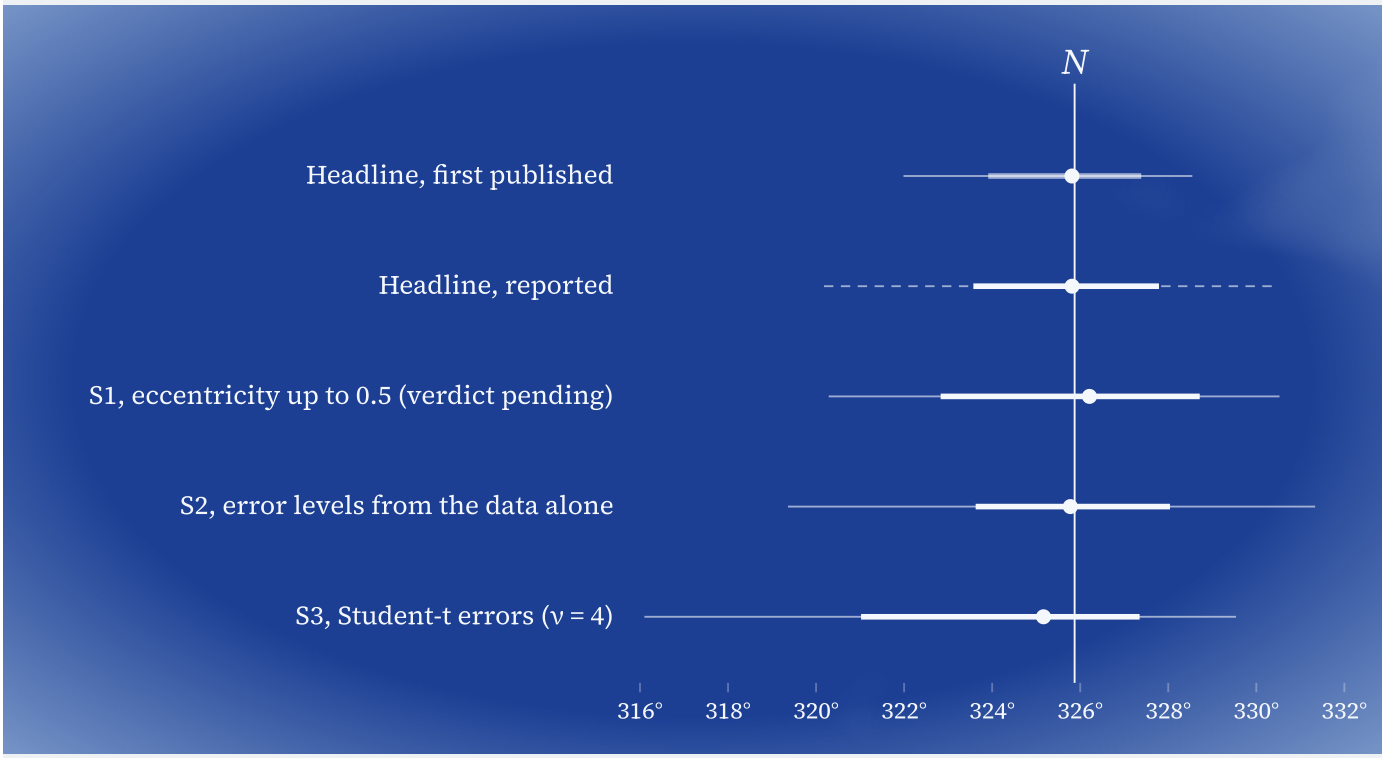


FIG. 6. The planet's geocentric longitude on Galle's night (ecliptic of date): the headline as first published (superseded) and as reported, and the three sensitivity runs. Dot: median; thick line: 68 % range; thin line: 95 % range, dashed where provisional. *N* Neptune (JPL Horizons). Reading rules: docs/notes/M6.md, M6c sensitivity runs S1–S3: pre-registration (2981aa1).

§ 6

Limitations

- Transcription (A6a): 0 of 28 rows in error; 95 % upper bound on the per-row error rate 10.1 % (exact one-sided Clopper–Pearson, 0 of 28); the remaining 251 rows are unchecked. The transcription was machine-assisted, not done by an independent human.
- Reduction check against JPL (A6b): comparison against per-group precision not made: no precision figure was fixed before the data were seen. Medians reported (Table 1).
- The 95 % range is provisional: M8b did not meet the replacement rule (std_95_bounds_le_0p4). The 68 % range is the headline.
- The S1 verdict is pending: the published run and M8's faster likelihood disagree, and a full-likelihood arbiter run decides.
- Error model: the evidence favours Student-t errors (S3, ln B = +13.7), but the Gaussian error model was fixed by the pre-registration and is kept. S3 shows the result is insensitive to it: the median moves 0.37 headline half-widths (§ 5).
- Diurnal parallax is not modelled (up to 0.5″).

- The headline plan was first pushed after its run had launched; the logs cannot rule out that one of its result files was opened by hand during the run (§ 3.1).
- Plate II and Fig. 5 are diagnostics made after the fact from the pre-registered posterior; they were not pre-registered.

§ 7

Data availability

The transcription of Table 77 (279 rows, with both passes, the resolution log and the misprint table) is on Zenodo under CC0 1.0. Cite the first DOI for the dataset in general and the second for exact reproduction, and cite Le Verrier (1846) for the observations themselves.

All versions 10.5281/zenodo.24044791

Pahadiya, A. (2026). Le Verrier (1846), Table n° 77: 279 observations of Uranus, 1690–1845, transcribed and reconciled [Data set]. Zenodo. https://doi.org/10.5281/zenodo.24044791

Version 1.0.0 10.5281/zenodo.24044792

Pahadiya, A. (2026). Le Verrier (1846), Table n° 77: 279 observations of Uranus, 1690–1845, transcribed and reconciled (Version 1.0.0) [Data set]. Zenodo. https://doi.org/10.5281/zenodo.24044792

The code and documentation are at github.com/atishaypahadiya/neptune-fork under the MIT License; the files in the repository's data/historical/ folder, which hold the transcription, are under CC0 1.0. Every number on this page is read from the site's data files, which the repository's scripts regenerate.

§ 8

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